

QIXUN WANG

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Education

Peking University

Ph.D. Student

Sep. 2023 – Current

School of Intelligence Science and Technology

Peking University

B.S.

Sep. 2019 – July 2023

School of Electronics Engineering and Computer Science

Research Interests

Reasoning of Multimodal Large Language Models (MLLMs): Enhancing the reasoning and perception capabilities of MLLMs in real-world multimodal scenarios.

Out-of-distribution (OOD) Generalization: Theoretical understanding and algorithm design for OOD robustness in computer vision, graph learning, and in-context learning of LLMs.

Selected Publications

Monet: Reasoning in Latent Visual Space Beyond Images and Language

Arxiv Preprint 2025

- Qixun Wang, Yang shi, Yifei Wang, Yuanxing Zhang, Pengfei Wan, Kun Gai, Xianghua Ying, Yisen Wang
- Propose a new SFT + RL training framework for multimodal latent reasoning, achieving significant improvements on both in-domain and OOD visual reasoning benchmarks.

Can In-context Learning Really Generalize to Out-of-distribution Tasks?

ICLR 2025

- Qixun Wang, Yifei Wang, Xianghua Ying, Yisen Wang
- Reveal the capability boundary and algorithm selection mechanism of ICL on OOD tasks through carefully designed experiments and theoretical analysis.

Dissecting the Failure of Invariant Learning on Graphs

NeurIPS 2024

- Qixun Wang, Yifei Wang, Yisen Wang, Xianghua Ying
- Theoretically and empirically demonstrate the failure modes of classic invariant learning approaches on graph data, and propose a new training objective with significant performance gains and theoretical guarantees.

Improving Out-of-distribution Robustness by Adversarial Training with Structured Priors

NeurIPS 2022

- Qixun Wang*, Yifei Wang*, Yisen Wang, Xianghua Ying
- Propose a simple yet effective low-rank adversarial training strategy that improves the OOD generalization of visual recognition models.

Technical Skills

Languages: Python, C++, shell

Frameworks: PyTorch, Transformers, vLLM, verl, DeepSpeed, TRL

Awards

- The Third-Class Scholarship of Peking University, 2025
- Merit Student at Peking University, 2025
- Outstanding Graduate of Peking University, 2023
- Yanchuang Capital Scholarship (Top 6%), 2022
- Merit Student at Peking University (Top 6%) 2022
- Academic Innovation Award at Peking University (Top 1%), 2022
- Award for Academic Excellence, 2021